

Foundations of Artificial Intelligence
Practice Exam Spring Semester 2026

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Name: _____

Matriculation number: _____

- The exam consists of a **multiple-choice part** and 6 **additional questions**.
- Please **write your name and your matriculation number on this title sheet**.
- You may prepare and use **one sheet of A4 paper with notes** (using both sides). Other aids such as lecture slides, notes, books, or calculators are not allowed. All electronic devices (such as mobile phones, smartwatches or earbuds) must be switched off.
- Please use a **blue or black pen** to answer the question, no red pen or pencil.
- You have **5 minutes** to read the exam. In this time, it is not allowed to use a pen.
- Afterwards, you have **105 minutes** for working on the exam.
- For answering the questions, please use the space directly below each question. If you require more space, please use the space on the next sheet.
- In case you develop several partial solutions for a question, please indicate clearly which one should be marked.

	Possible marks	Marks achieved
Multiple-Choice Questions	20	
Question 1	10	
Question 2	10	
Question 3	10	
Question 4	10	
Question 5	10	
Question 6	10	
Total	80	
Grade	(1.0–6.0)	

Multiple-Choice Questions (10×2 marks)

Note: in every block, any number of answers (0–4) may be correct. Ticking an incorrect box or not ticking a correct box is a mistake. Each block is marked separately and gives 2 marks if there are no mistakes, 1 mark if there is one mistake, and 0 marks if there are two or more mistakes.

- (a) Which of the following statements correctly describe differences between tree search and graph search?
- ☐ Graph search uses a closed list; tree search does not.
 - ☐ Graph search uses an open list; tree search does not.
 - ☐ Graph search is usually complete; this is not true for tree search.
 - ☐ Graph search is performed depth-first; tree search is performed breadth-first.
- (b) Which of these can be infinite in a state space (as defined in the course)?
- ☐ number of states
 - ☐ number of paths
 - ☐ number of solutions
 - ☐ number of actions
- (c) Which of the following statements about heuristics are true?
- ☐ The perfect heuristic h^* is safe.
 - ☐ Every admissible heuristic is safe.
 - ☐ Every safe heuristic is admissible.
 - ☐ Every consistent heuristic is safe.
- (d) Which of the following statements about local search algorithms for combinatorial optimization are true?
- ☐ If there exists a solution, hill-climbing will find it.
 - ☐ If there exists no solution, hill-climbing will prove that this is the case.
 - ☐ Local optima are a major challenge for local search algorithms.
 - ☐ Making random steps can help a local search algorithm escape local optima.
- (e) Which of the following statements about constraint satisfaction algorithms are true?
- ☐ *BacktrackingWithInference* usually takes more time per recursive function call than *NaiveBacktracking*, but often requires much fewer recursive function calls.
 - ☐ A good value ordering can help escape unsolvable parts of the search tree as quickly as possible.
 - ☐ After *NaiveBacktracking* finds an initial solution, it then continues searching for better solutions until it has exhausted all possibilities.
 - ☐ On a constraint network with n variables with domain size k , the runtime of *NaiveBacktracking* is $\Theta(k^n)$ both in the best case and in the worst case.

- (f) Which of the following statements about inference methods for constraint networks are true?
- ☐ It is more expensive to make a constraint network arc-consistent than to perform forward checking.
 - ☐ It is more expensive to make a constraint network path-consistent than to make it arc-consistent.
 - ☐ The AC-3 algorithm reduces unnecessary work by keeping track of which variable pairs might violate arc consistency.
 - ☐ The AC-1 algorithm has a better worst-case runtime than the AC-3 algorithm.
- (g) Which of the following are models of the formula $(X \wedge \neg(Y \wedge \neg Z))$?
- ☐ $\{X \mapsto \mathbf{T}, Y \mapsto \mathbf{T}, Z \mapsto \mathbf{T}\}$
 - ☐ $\{X \mapsto \mathbf{T}, Y \mapsto \mathbf{F}, Z \mapsto \mathbf{F}\}$
 - ☐ $\{X \mapsto \mathbf{F}, Y \mapsto \mathbf{F}, Z \mapsto \mathbf{F}\}$
 - ☐ $(X \vee \neg X)$
- (h) Which of the following statements about propositional formulas are true?
- ☐ Every valid formula is satisfiable.
 - ☐ Every unsatisfiable formula is falsifiable.
 - ☐ A formula can be satisfiable and falsifiable.
 - ☐ If φ is satisfiable, then $\neg\varphi$ must be unsatisfiable.
- (i) Which of the following statements about automated planning heuristics are true?
- ☐ The FF heuristic is an admissible variant of the h^{add} heuristic.
 - ☐ The abstraction heuristic h^α is admissible for every abstraction function α .
 - ☐ All pattern database heuristics are admissible.
 - ☐ All pattern database heuristics are abstraction heuristics.
- (j) Which statements about Monte-Carlo tree search are correct?
- ☐ Monte-Carlo tree search requires a heuristic.
 - ☐ Every iteration of Monte-Carlo tree search adds at most 1 node to the search tree.
 - ☐ With the UCB1 tree policy, the final strategy computed by the algorithm does not depend on which default policy is used (if we run the algorithm long enough).
 - ☐ All variants of Monte-Carlo tree search converge to an optimal strategy.

Question 1 (blind search, 2+2+2+2+2 marks)

For each of the five parts, give an example of a state space with *exactly four states* with the given property, or explain briefly (1–2 sentences) why such a state space does not exist. If you give an example, you do not need to explain why it has the given property. The example must work regardless of tie-breaking (the order in which *succ* returns the successors).

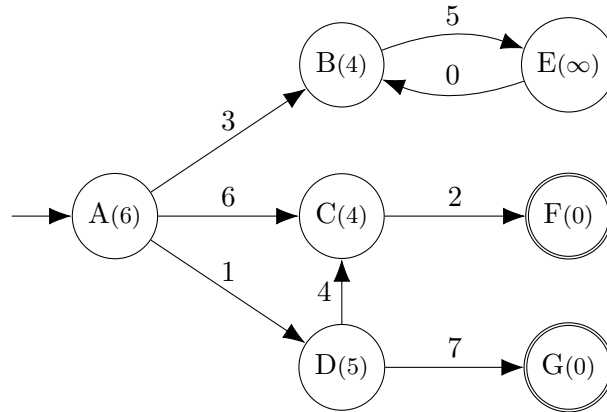
It is sufficient to show the state spaces visually. Make sure that the initial state, goal states, direction of state transitions and action costs (if they are important for the answer) are clearly shown. Your example state spaces may be solvable or unsolvable.

- (a) a state space where BFS-Tree fails to terminate
- (b) a state space where BFS-Graph returns an optimal solution
- (c) a state space where BFS-Graph returns a non-optimal solution
- (d) a state space where uniform cost search fails to terminate
- (e) a state space where iterative deepening depth-first search returns a non-optimal solution

Additional space for question 1:

Question 2 (heuristic search, 2+2+4+2 marks)

Consider the following state space with initial state A and goal states F and G . Transitions are labeled directly with their cost rather than denoting actions and their costs separately. Numbers in parentheses inside the states denote values of heuristic h .



- Is h safe? Briefly justify your answer.
- Is h admissible? Briefly justify your answer.
- List the sequence of expansions performed by A^* on this state space. The final iteration in which the solution is extracted counts as an expansion for this question. It is sufficient to describe each expansion by giving the *state* of the expanded node along with its f , g and h values. For example, if a node with state Z is expanded with f value 10, g value 3 and h value 7, it suffices to write “ Z : $10 = 3 + 7$ ”.
- Give the sequence of states in the solution found by A^* , and give the cost of the solution. Is the solution optimal? If not, state a cheaper solution and its cost.

Additional space for question 2:

Question 3 (constraint satisfaction, 4.5+1+1+1+2.5)

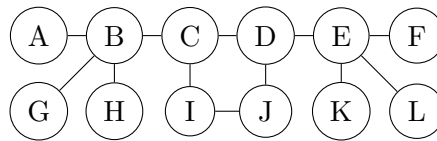
For parts (a)–(d), consider a constraint network with variables $\{u, v, w\}$, all with domain $\{1, 2, 3, 4\}$. The constraints are given by:

- $R_{uv} = \{\langle u, v \rangle \mid u \leq v\}$
- $R_{uw} = \{\langle u, w \rangle \mid u + w \leq 3\}$
- $R_{vw} = \{\langle 1, 2 \rangle, \langle 1, 4 \rangle, \langle 2, 1 \rangle, \langle 3, 1 \rangle, \langle 4, 4 \rangle\}$

- (a) For each constraint R_{xy} , state whether x is arc-consistent with respect to y and whether y is arc-consistent with respect to x . Whenever the answer to one of these questions is “no”, state which values can be pruned from which domains by enforcing arc consistency.

Consider each constraint separately, i.e., on the *original* constraint network. (For example, when considering the third constraint, do not revise the domains according to the first two constraints first.)

- (b) Give a solution to the constraint network.
- (c) Give a total assignment for this constraint network that isn’t a solution and state which of the three constraints above it violates. (If it violates multiple constraints, state all.)
- (d) How many total assignments for this constraint network exist?
- (e) Consider a constraint network with the following constraint graph. If you use the *BacktrackingWithInference* algorithm and enforce arc consistency at each inference step, which is the best variable to consider first in the variable order and why? (If there are multiple equally good first variables, you may name any of them.)



Additional space for question 3:

Question 4 (propositional logic, 1+2+2+5 marks)

For parts (a)–(c), you do not need to justify your answer.

- (a) Give an example of an unsatisfiable propositional formula in conjunctive normal form.
- (b) Give an example of a propositional formula that is neither in conjunctive normal form nor in disjunctive normal form, using only atomic formulas, conjunctions and disjunctions. (Do not use the symbols $\perp$, $\top$, $\neg$, $\rightarrow$ or $\leftrightarrow$.)
- (c) List all models of the formula $(P \rightarrow \neg Q)$ over the alphabet $\{P, Q\}$. Use the correct formal notation to state the models; do not just list a truth table.
- (d) Use DPLL to find a satisfying assignment for

$$\Delta = \{\{P, \neg Q, R\}, \{P, \neg Q, \neg R\}, \{\neg P\}, \{\neg P, R\}, \{\neg Q, R\}, \{Q, \neg R\}\}$$

or show that the formula is unsatisfiable. Clearly state the result computed by the algorithm (the found model or the result “unsatisfiable”).

When you can choose a variable, prefer P over Q and Q over R . When you must consider both truth values in some order, try **T** before **F**. Clearly show the recursive call structure (for examples as a tree). For each recursive call, specify the current set of clauses and which rule of the algorithm is used (empty clause, no clauses, unit propagation, or splitting rule).

Additional space for question 4:

Question 5 (automated planning, 2+2+2+2+2 marks)

Consider the STRIPS planning task $\Pi = \langle V, I, G, A \rangle$ with

$$V = \{p, q, r\}$$

$$I = \{p\}$$

$$G = \{p, q, r\}$$

$$A = \{a_1, a_2, a_3\}, \text{ where}$$

$$pre(a_1) = \{p\}, \quad add(a_1) = \{q\}, \quad del(a_1) = \emptyset, \quad cost(a_1) = 1$$

$$pre(a_2) = \{q\}, \quad add(a_2) = \{r\}, \quad del(a_2) = \{q\}, \quad cost(a_2) = 10$$

$$pre(a_3) = \{p, q\}, \quad add(a_3) = \{r\}, \quad del(a_3) = \{p\}, \quad cost(a_3) = 1.$$

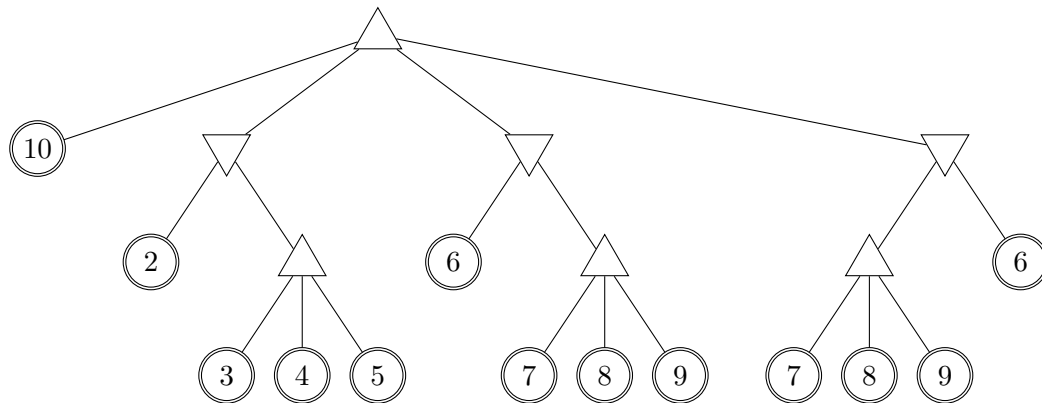
- (a) How many states does Π have? (*Note:* all states, not just reachable states.)
- (b) Give an example of a state in which action a_3 is not applicable.
- (c) Give an example of a transition in the state space $\mathcal{S}(\Pi)$.
- (d) State an optimal solution for Π and its cost.
- (e) State an optimal solution for Π^+ (the delete relaxation of Π) and its cost.

You do not need to justify your answers.

Additional space for question 5:

Question 6 (board games, 5+3+2 marks)

Consider the following game tree for a two-player game:



Reminder: Triangles point up for MAX nodes and point down for MIN nodes.

- Compute the minimax values of all non-terminal positions and add them to the figure. Mark the move played by MAX or MIN in all non-terminal positions.
- Mark all positions (terminal or non-terminal) that are cut off by Alpha-Beta search. Moves are considered in left-to-right order.
- At which single node and how would you change the move ordering to maximize the number of cutoffs in Alpha-Beta search? You do not need to justify your answer.

Additional space for question 6:

